UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Milestone 4: Electrical Conduction Around the Electrode

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Input status: Electrode radius and the two tissue conductivities come from the handout. Outer reference radius, applied voltage, grid, contact-depth interpretation, and power-density threshold are configurable modeling assumptions, not official specifications.

Abstract

This report implements a steady finite-volume electrical-conduction model around a spherical catheter electrode at a myocardium–blood interface. It calculates voltage, electric field, current density, Joule power density, electrode current, impedance, and regional power. Current conservation, power balance, analytical resistance, and mesh refinement are used as verification checks. The results are reproducible for the stated inputs, but the outer boundary and drive voltage are illustrative, and a power-density contour is not a clinical lesion prediction.

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1 Objective and input status

Milestone 4 models how electrical energy flows from the catheter electrode through blood and myocardium. The handout inputs available for this model are a spherical electrode radius $a = 2\,\mathrm{mm}$, myocardium conductivity $\sigma_{\mathrm{myo}} = 0.54\,\mathrm{S\,m^{-1}}$, blood conductivity $\sigma_{\mathrm{blood}} = 0.70\,\mathrm{S\,m^{-1}}$, and a contact-depth range from 0 to 4 mm. The model is static, so it solves for steady current rather than a time-varying waveform.

Table 1: Input values and their status.

Input	Value	Status
Electrode radius	2 mm	Handout
Myocardium conductivity	$0.54\,\mathrm{S\,m^{-1}}$	Handout
Blood conductivity	$0.70\,\mathrm{S\,m^{-1}}$	Handout
Contact-depth range	0 mm to 4 mm	Handout range; convention interpreted
Outer reference radius	20 mm	Illustrative, configurable
Electrode voltage	1 V	Illustrative, configurable
Power-density threshold	$5.0 \times 10^4\,\mathrm{W\,m^{-3}}$	Illustrative proxy only

The outer reference radius and applied voltage were not specified in the available starter materials. They are named parameters in the solver. The 20 mm radius gives a useful comparison with the handout's approximate $58\,\Omega$ reference, but that agreement does not make the assumed boundary an official value. For a linear conduction model, changing the drive voltage scales current and power but does not change impedance.

Contact depth is defined as insertion from first touch. In coordinates centered on the electrode, the planar tissue interface is at $z = d - a$. Thus 0 mm is tangent contact, 2 mm places the electrode center at the interface, and 4 mm places the sphere fully in myocardium. This geometric interpretation should be checked against the instructor's intended definition.

2 Governing equations

For static conduction, charge conservation and Ohm's law give

$$\nabla \cdot (\sigma \nabla V) = 0, \qquad \mathbf{E} = -\nabla V, \qquad \mathbf{J} = \sigma \mathbf{E}.$$

The local Joule heating density is

$$q = \mathbf{J} \cdot \mathbf{E} = \sigma |\mathbf{E}|^2 \quad [\mathrm{W\,m^{-3}}].$$

The finite-volume current through an electrode surface is summed over its boundary faces. The input impedance and power are

$$R_{\mathrm{in}} = \frac{V_{\mathrm{electrode}}}{I_{\mathrm{electrode}}}, \qquad P_{\mathrm{in}} = V_{\mathrm{electrode}} I_{\mathrm{electrode}}.$$

3 Finite-volume implementation

The computational domain is a sphere of radius 20 mm, centered on the electrode and held at zero potential. The electrode surface is held at 1 V. A cell-centered Cartesian grid marks cells inside the electrode and outside the outer sphere as fixed-potential boundaries; remaining cells are tissue unknowns. The planar interface assigns each unknown cell to myocardium or blood by its center position.

For neighboring tissue cells, the conductance between them is

$$G_f = \sigma_f \frac{A_f}{\Delta x},$$

where A_f is the face area and Δx is cell spacing. The face conductivity σ_f uses the harmonic mean of the two neighboring conductivities. The resulting sparse system is solved by conjugate gradients. The spherical boundaries and interface are voxelized on the Cartesian mesh. Local field and Joule-density values are reconstructed from averaged face fields; regional powers are summed from finite-volume face dissipation.

For a centered electrode at the interface, each tissue region occupies one hemisphere. The analytical spherical-resistance comparison is

$$R_{\text{analytic}} = \frac{1/a - 1/R_{\text{out}}}{2\pi(\sigma_{\text{myo}} + \sigma_{\text{blood}})}.$$

Using $a = 2\,\text{mm}$, $R_{\text{out}} = 20\,\text{mm}$, and the two conductivities gives $57.757\,842\,\Omega$.

4 Centered-interface results and verification

At the illustrative 1 V drive, the finest sampled grid (6 cells per electrode radius) gives $R_{\text{in}} = 57.217\,072\,\Omega$ and $I_{\text{electrode}} = 17.477\,301\,\text{mA}$. Input power is $17.477\,301\,\text{mW}$. The regional finite-volume power is $7.611\,083\,\text{mW}$ in myocardium and $9.866\,218\,\text{mW}$ in blood, so 43.55% of the dissipated power is in myocardium.

Table 2: Centered-interface mesh refinement compared with the analytical resistance.

Cells per radius	Unknown cells	CG iterations	Resistance (Ω)	Error (%)
2	33,520	89	59.864220	3.647
4	267,816	167	56.576475	2.045
6	904,048	235	57.217072	0.936

The discretized resistance error does not decrease at every refinement step because a voxelized sphere changes its staircase shape as the grid changes. The finest sampled result is closer to the analytical value than the coarsest. The computed 57.217072 Ω is 0.936% below the analytical result and 1.35% below the handout's approximate 58 Ω comparison.

Table 3: Finite-volume conservation and solver checks at 6 cells per radius.

Check	Result	Outcome
Electrode current	17.477 301 mA	
Outer-boundary current	17.477 301 mA	PASS
Current-balance relative error	$4.14 \times 10^{-7}\%$	PASS
Input power	17.477 301 mW	
Sum of regional face powers	17.477 301 mW	PASS
Power-balance relative error	$3.68 \times 10^{-8}\%$	PASS
Conjugate-gradient solve	235 iterations, converged	PASS

The arithmetic-versus-harmonic face-averaging comparison at 1 mm contact depth differs by 0.020% in impedance. Harmonic averaging is used for the reported results because it represents conduction through two materials in series across a face.

5 Contact-depth study

The next table uses a 3-cells-per-radius grid and the same illustrative 1 V drive at each depth. The last column counts myocardium cells above the illustrative power-density threshold. It is a numerical threshold proxy, not a biological lesion volume.

Table 4: Calculated contact-depth sensitivity.

Depth (mm)	Resistance (Ω)	Current (mA)	Myocardium power share (%)	Thresholded volume (mm^3)
0	50.69360	19.726	21.15	3.556
1	51.67102	19.353	27.73	9.481
2	53.95999	18.532	43.55	21.333
3	55.24242	18.102	52.71	29.630
4	57.92809	17.263	72.47	47.407

As the electrode moves farther into myocardium, the computed myocardium share of power increases in this model. At fixed voltage, the total current and power decline across the sweep, while the thresholded proxy volume increases. The result is sensitive to the assumed threshold, contact definition, and material interface placement.

6 Figures

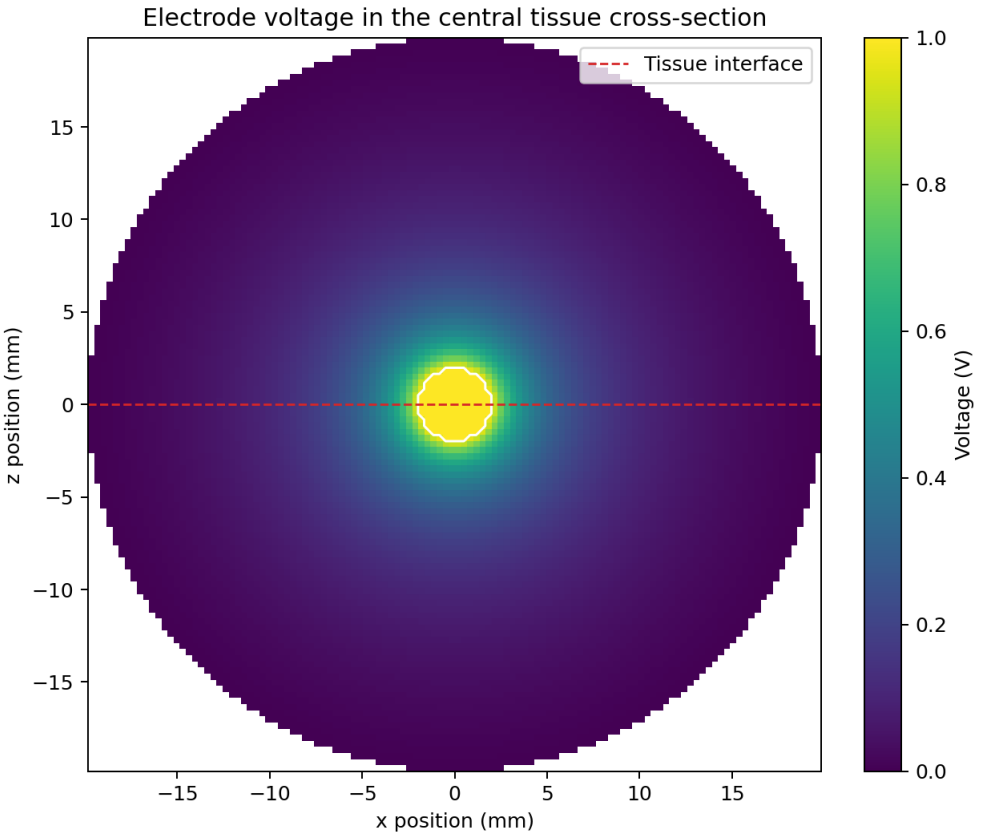


Figure 1: Voltage on the central x-z plane at $y = 0$. The dashed line is the myocardium–blood interface; the white outline marks the spherical electrode.

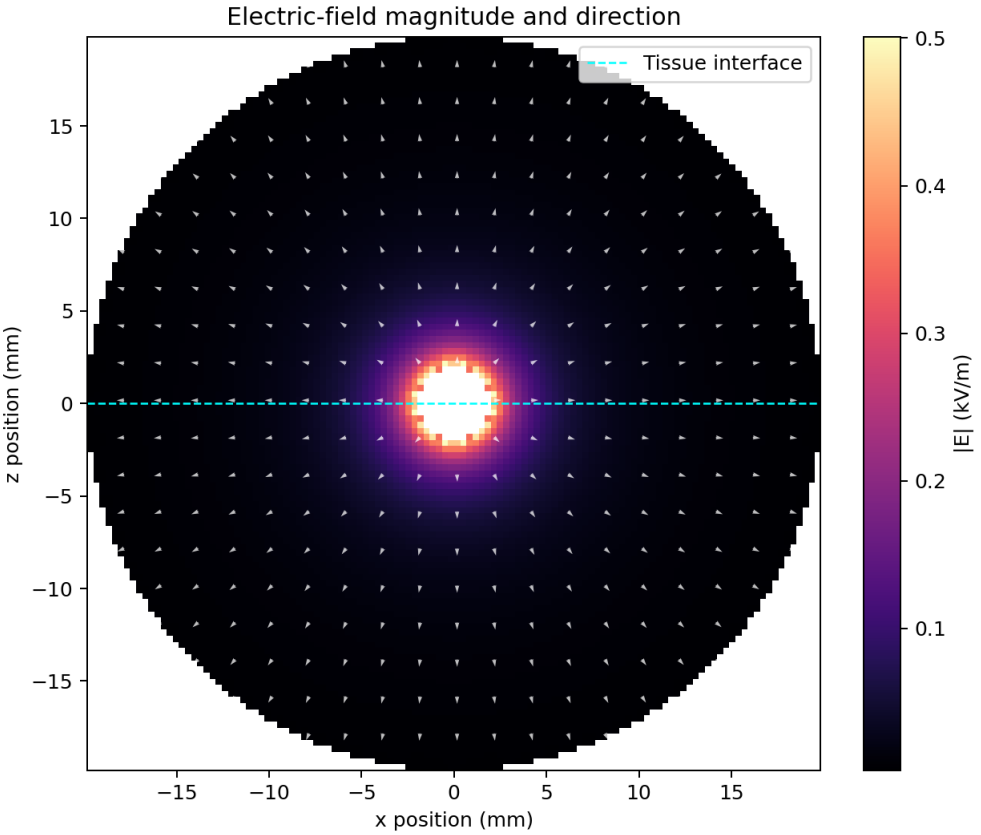


Figure 2: Electric-field magnitude and direction on the same central plane. The field points away from the driven electrode; its strength is greatest near the electrode surface.

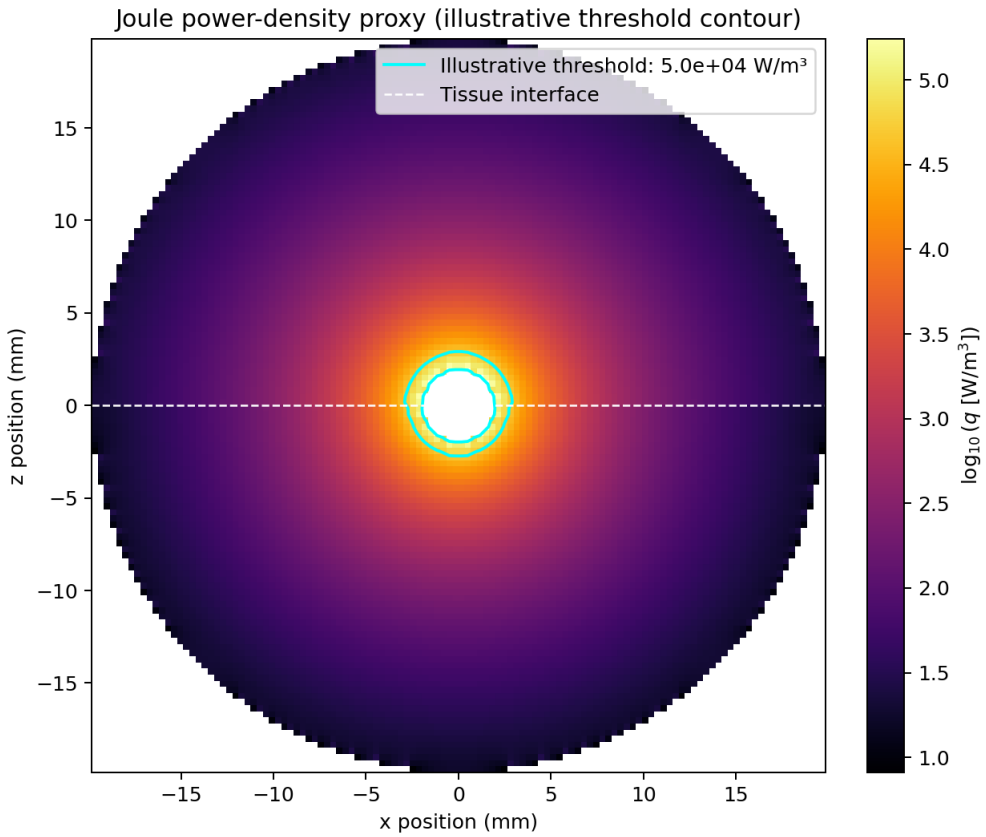


Figure 3: Logarithmic Joule power density and an illustrative $5.0 \times 10^4\,\mathrm{W\,m^{-3}}$ contour. This threshold is a plotting assumption and is not a validated lesion boundary.

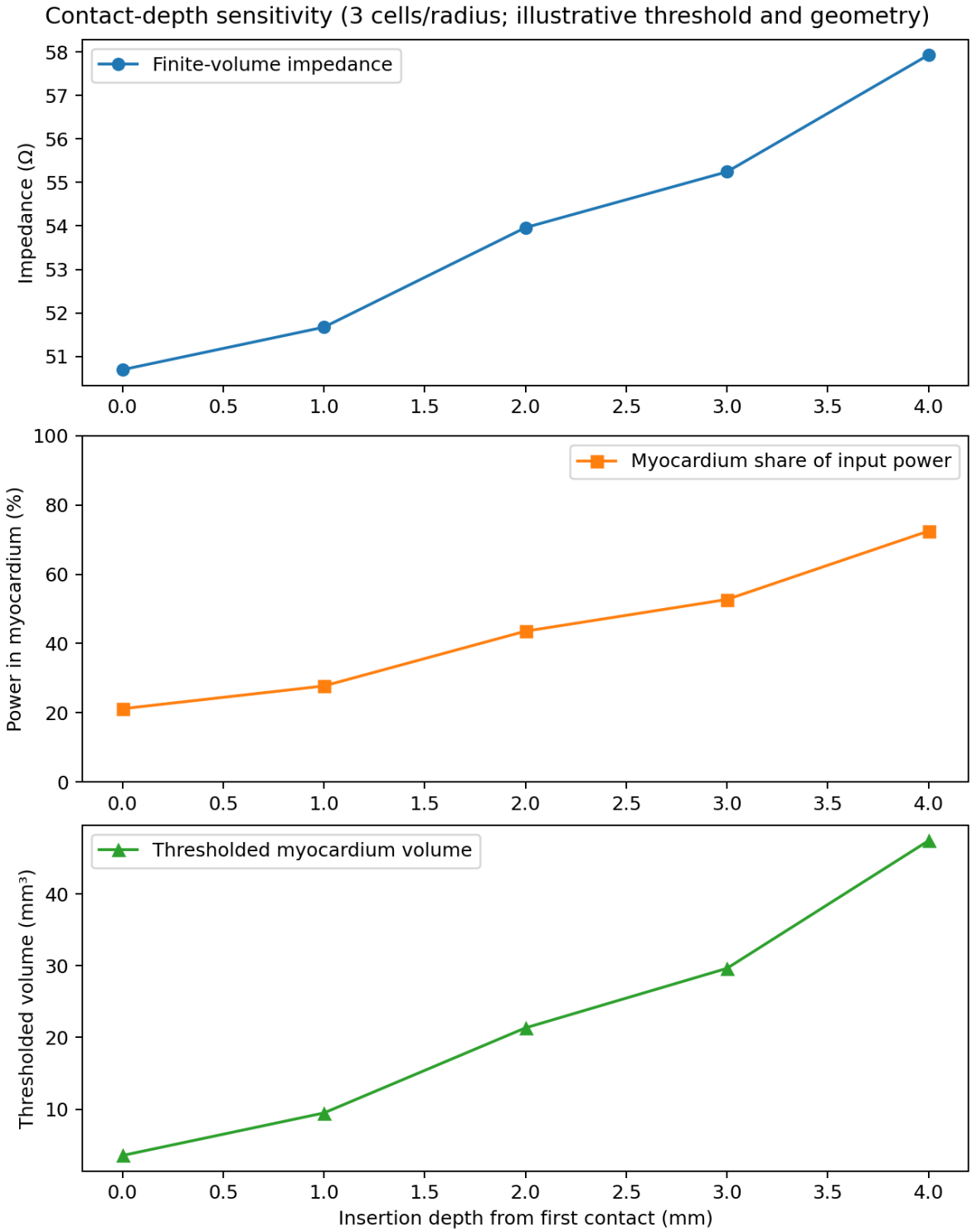


Figure 4: Impedance, myocardium share of power, and thresholded myocardium volume versus insertion depth. The last curve is only an illustrative power-density-threshold proxy.

7 Engineering interpretation and limitations

The model's current and power balances are satisfied to far below 0.1%, and the centered-interface resistance approaches the analytical comparison as the mesh is refined overall. The depth sweep suggests that electrode location can substantially redistribute power between myocardium and blood. The reported values are not a complete treatment-planning or safety analysis.

The outer reference radius, voltage, and power-density threshold are illustrative, while the depth convention needs confirmation. The model uses constant conductivities and a planar interface, with fixed potentials at the electrode and outer sphere. It does not include electrode contact impedance, time-varying stimulation, temperature-dependent conductivity, bioheat transfer, tissue damage kinetics, or clinical lesion validation. Official conductivity values, boundary conditions, operating waveform, and contact geometry should be confirmed before using these results for design. No professor-provided Milestone 4 acceptance tests were present. The checks in this report are model self-checks, not official acceptance tests.

8 How I Would Explain Milestone 4 to My Professor

I divided the tissue volume into small cells and used charge conservation at each cell to solve for voltage. The conductivity of each tissue region determines how easily current crosses each cell face. From the voltage solution I calculated electric field, current density, and Joule heating. I then checked that the current leaving the electrode reaches the outer boundary and that the sum of the computed regional power agrees with electrical input power. For a centered electrode I also compared the numerical resistance with the spherical analytical result. The trends are useful for explaining how insertion depth redistributes power, but the outer boundary, voltage, and threshold are illustrative, so these results are not a clinical lesion prediction.